

4. PROJECT DESIGN PHASE

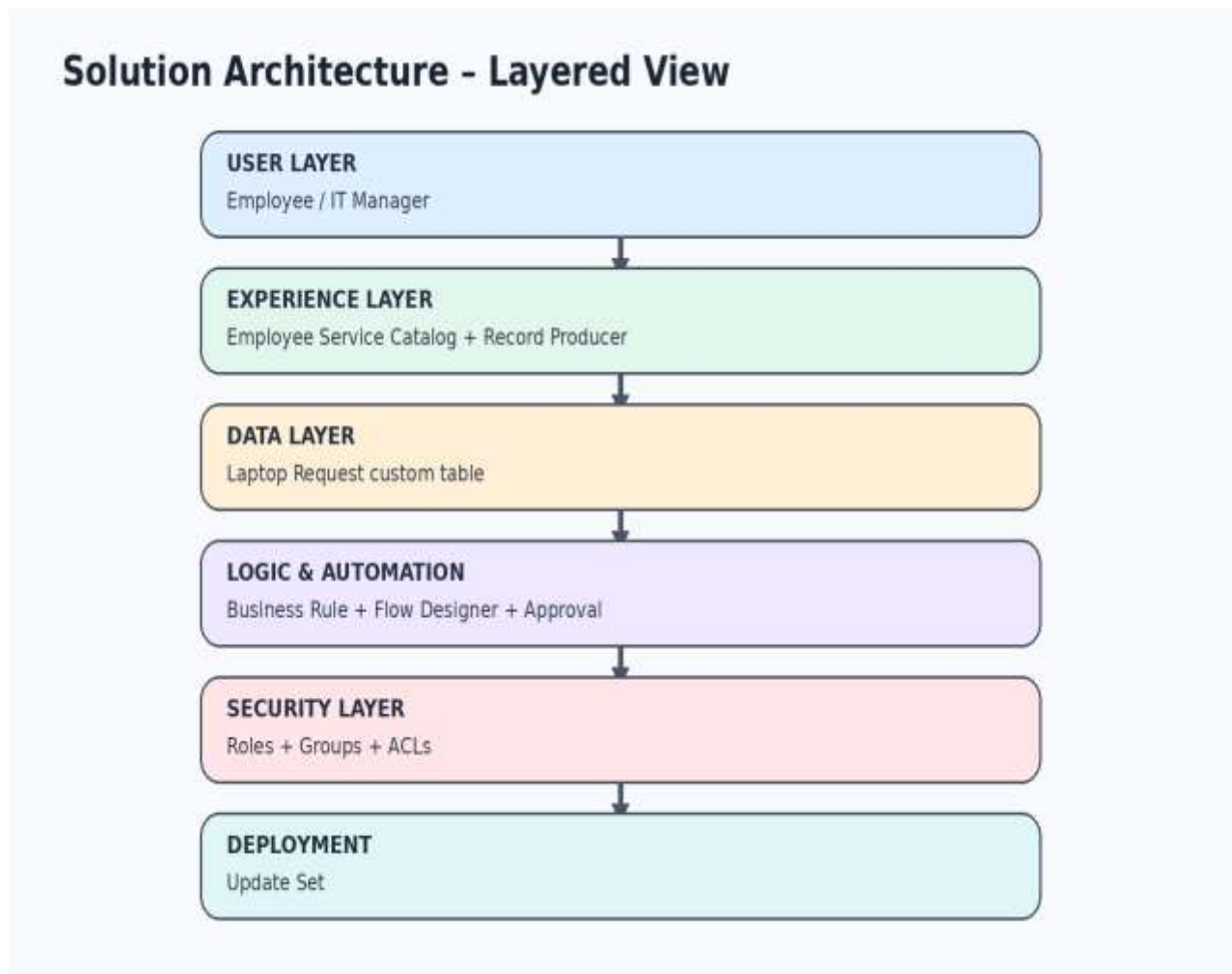
ServiceNow-Based Employee Laptop Request Application

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4.1 Design Objective

The design phase defines how the approved requirements are represented within ServiceNow. The design connects the employee-facing request experience with the data model, backend validation, automation, approval and security layers.

4.2 Solution Architecture



4.3 Application Component Design

Component	Design Purpose
Custom Scoped Application	Provides a dedicated application scope for project artifacts.
Laptop Request Table	Acts as the main data source for employee laptop requests.
Record Producer	Provides the employee-facing request interface.
Business Rule	Validates the one-request-per-year policy before insert.
Flow Designer	Automates approval, decision handling and notifications.
Approval Relationship	Connects the request with the ServiceNow approval table.
Roles / Groups	Organize access according to responsibility.
ACLs	Control read, create and write operations.
Update Set	Captures configuration changes for deployment.

4.4 Data Model Design

Field Group	Design
Identification	Number – Auto Number
Employee	Requester – Reference; Employee Email – Email
Request Details	Request Date – Date/Time; Short Description – String
Approval	Manager – Reference
Classification	Type of Request – Basic / Advanced; Reason – Software Development / Personal Use / Other
Lifecycle	Status – Pending Approval / Approved / Rejected

4.5 Record Producer Design

The Record Producer named Laptop Request is mapped to the Laptop Request table. Variables are configured according to the table field types and mapped to the corresponding fields. The producer is added to the Employee Service Catalog so employees have a clear front-end entry point.

4.6 Business Rule Design

Property	Design Decision
Name	Laptop Request BR
Table	Laptop Request
Execution	Before Insert
Purpose	Prevent a second laptop request by the same employee in the current calendar year.
Success	Allow the request to continue when no current-year duplicate exists.
Failure	Block the request and display an error message.

4.7 Automation Design

A new request starts the Flow Designer process. The request is initialized to Pending Approval, an approval is requested from the manager, and the result determines the final status and employee notification.

4.8 Security Design

Role	Read	Create	Write
it_employee	Yes	Yes	Yes
it_manager	Yes	No	No

4.9 Design Principle

The design separates user interaction, data storage, backend validation, automation and security. This makes the request lifecycle easier to understand, test and maintain without repeating the same configuration across multiple components.